

Lesson 5 Single PWM Servo Control

This section focuses on controlling the rotation of a single servo by utilizing the No.1 PWM interface on the expansion board.

1. Working Principle

In the program, modify the runtime and servo pulse width to control the rotation of the servo. The source code of the program is located in “/home/pi/board_demo/single_servos.py”.

```
29
30 if __name__ == '__main__':
31     while True:
32         board.pwm_servo_set_position(1, [[1, 1500]]) # 设置1号舵机脉宽为1500
33         0(set the pulse width of servo 1 to 1500)
34         time.sleep(1)
35         board.pwm_servo_set_position(1, [[1, 1000]]) # 设置1号舵机脉宽为1000
36         0(set the pulse width of servo 1 to 1000)
37         time.sleep(1)
38         board.pwm_servo_set_position(1, [[1, 500]]) # 设置1号舵机脉宽为500
39         0(set the pulse width of servo 1 to 500)
40         time.sleep(1)
41         board.pwm_servo_set_position(1, [[1, 1000]]) # 设置1号舵机脉宽为1000
42         0(set the pulse width of servo 1 to 1000)
43         time.sleep(1)
44         board.pwm_servo_set_position(1, [[1, 1500]]) # 设置1号舵机脉宽为1500
45         0(set the pulse width of servo 1 to 1500)
46         time.sleep(1)
47         board.pwm_servo_set_position(1, [[1, 2000]]) # 设置1号舵机脉宽为2000
48         0(set the pulse width of servo 1 to 2000)
49         time.sleep(1)
50         board.pwm_servo_set_position(1, [[1, 2500]]) # 设置1号舵机脉宽为2500
51         0(set the pulse width of servo 1 to 2500)
52         time.sleep(1)
53         if not start:
```

The program calls the “pwm_servo_set_position()” function in the “Board” library to implement the servo control. Take “board.pwm_servo_set_position(0.5, [[1, 1500]])” as an example, among them:

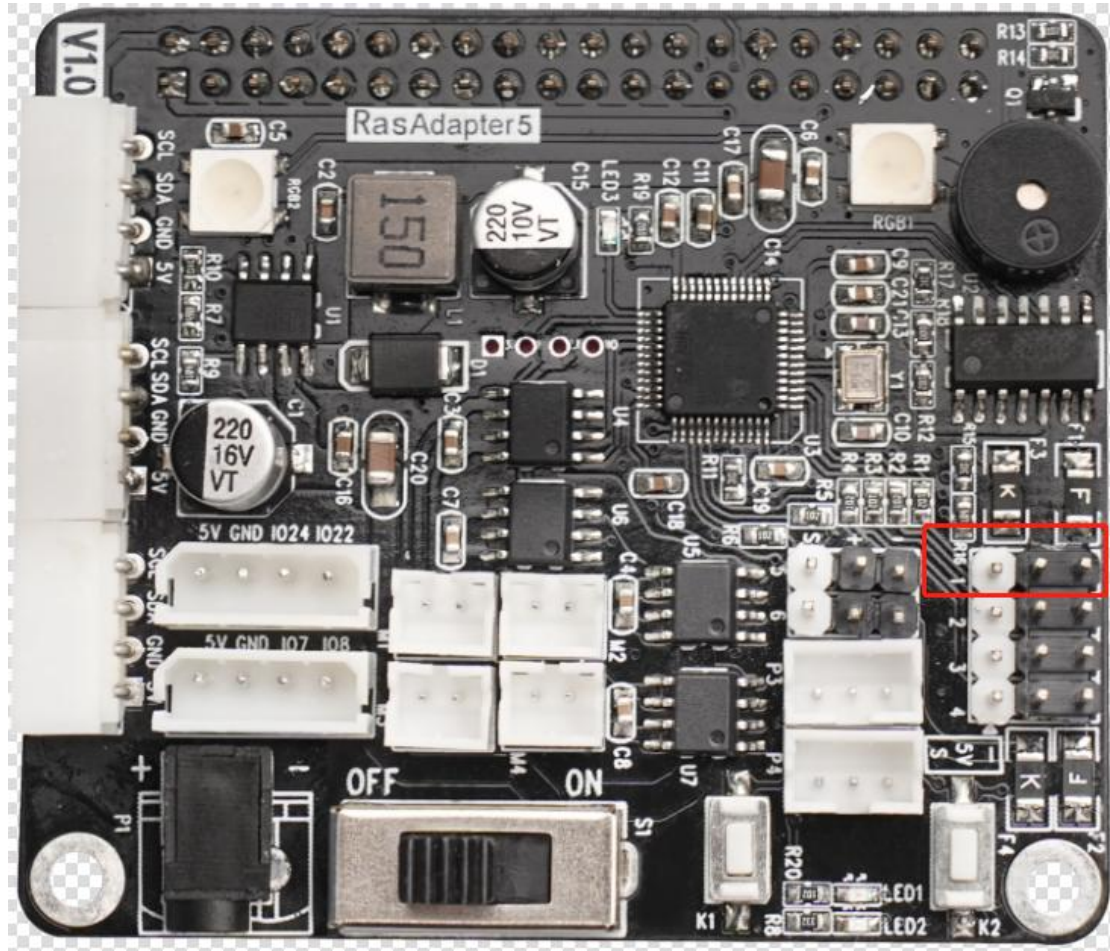
The first parameter “0.5” indicates the servo’s runtime is 500ms. If it is “1”, indicating the runtime is 1000ms.

The second parameter “[1, 1500]” represents setting the No.1 servo’s

pulse width as 1500.

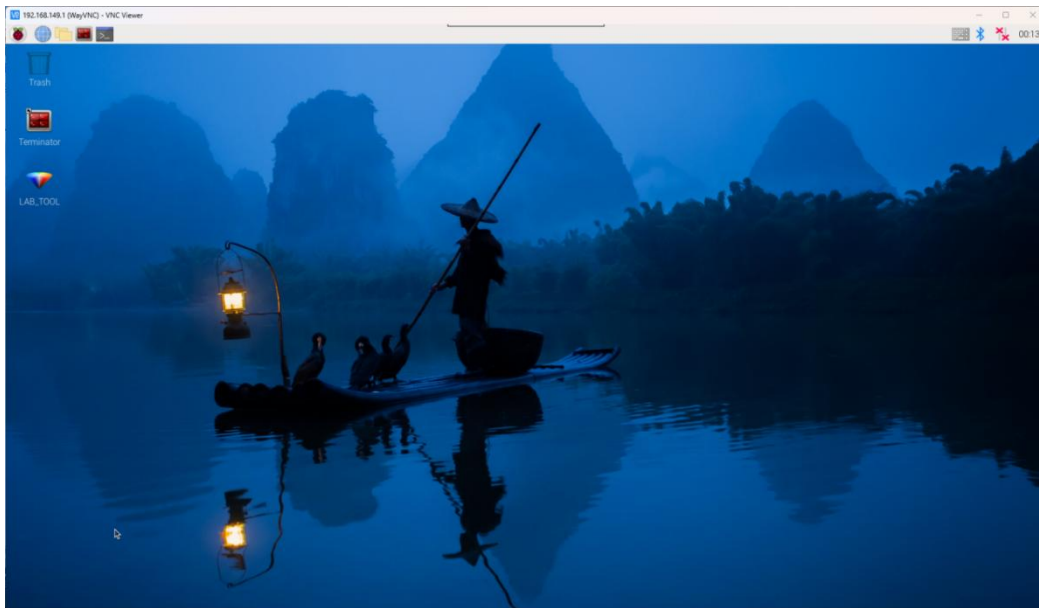
2. Getting Ready

There are 6 PWM servo interfaces on the Raspberry Pi expansion board. Let's test by connecting the No.1 servo, as shown below:

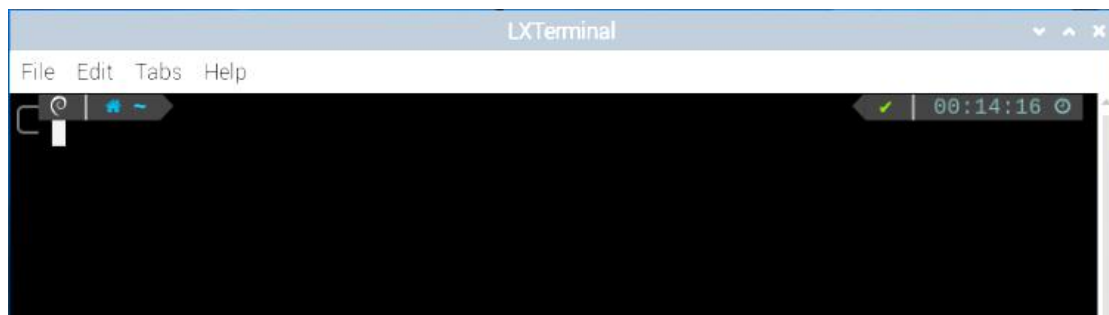


3. Operation Steps

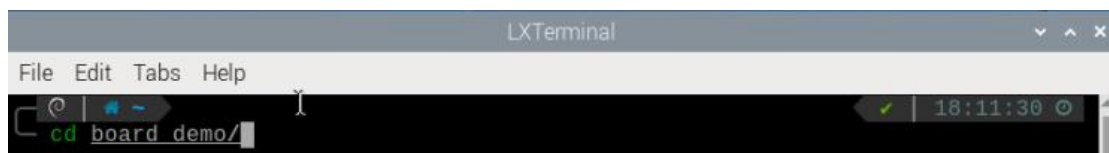
1) Turn the device on and connect the robot through the VNC remote connection tool.



2) Click  in the top left corner of the desktop, or press the shortcut key “Ctrl+Alt+T” to open the command line terminal.



3) Enter the command “cd board_demo/” to navigate to the directory where the demo program is located, then press “Enter”.



4) Enter the command of running the program “python3 single_servos.py”, and press “Enter”.

```
pi@raspberrypi:~/board_demo $ python3 single_servos.py
*****
*****功能:幻尔科技树莓派扩展板, 控制单个PWM舵机(function:Hiwonder Raspberry
Pi expansion board,  single PWM servo control)*****
*****
-----
Official website:https://www.hiwonder.com
Online mall:https://hiwonder.tmall.com
-----
Tips:
 * 按下Ctrl+C可关闭此次程序运行, 若失败请多次尝试! (Press "Ctrl+C" to exit the
program. If it fails, please try again multiple times!)
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```

5) To close the program, press “Ctrl+c”. If the program cannot be closed successfully, repeat this operation until it exits.

4. Program Outcome

Once the program is executed, the Raspberry Pi expansion board can adjust the pulse width of the servo to control its angular position, thereby achieving different motion effects.